



10.7.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to solving problems involving chords.

1. I can define a chord of a circle.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. I can define the relationship of angles and arcs formed by the intersection of two chords.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

3. I can solve problems involving chords of a circle.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.



Unit 10, Lesson 8: Solving Problems Involving Intersection of Secants



Benchmark

MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.



Learning Target

- ☐ I can solve problems involving secants drawn from a point external to the circle and their related angles.



10.8.1 Warm-Up: Satellite Operations Engineer

- Veronica Navarro is a satellite operations engineer. Before beginning her career, she received a bachelor's degree. One piece of advice she would give to high school students is to take as many classes in high school as possible because they are free. In her opinion, a degree beyond high school is needed in many careers because employers want to know a person can be trained and is capable of learning. Veronica's employer paid for her to receive additional degrees as a means to further develop her abilities and deepen her learning.

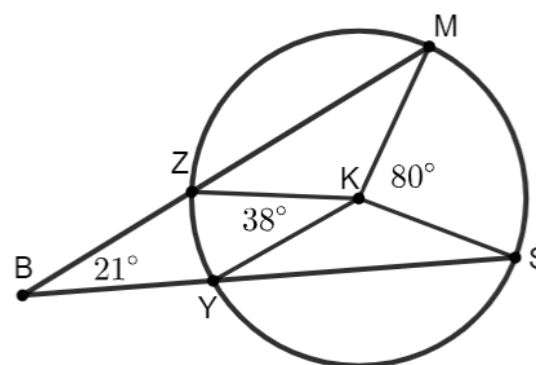
Why do you think it is important for companies to know you are capable of being trained?



10.8.2 Exploration: Angles Formed by Secants

A **secant** is a line or line segment that passes through two points on a circle. In the interior of the circle, a secant line segment forms a chord.

- $\angle MBS$ intersects Circle K , creating two intersecting arcs, where points M , Z , Y , and S are on the circle. The $m\angle ZKY = 38^\circ$, $m\angle SKM = 80^\circ$, and $m\angle MBS = 21^\circ$.



- Complete the table.

Arc	Central Angle Measure	Arc Measure
$\widehat{MS}$		
$\widehat{ZY}$		

- Add and subtract the two arc measures.

$$m\widehat{MS} + m\widehat{ZY} = \underline{\hspace{2cm}}$$

$$m\widehat{MS} - m\widehat{ZY} = \underline{\hspace{2cm}}$$

- c. Do the operations in Part B compare to the measure of $\angle MBS$? If so, explain.

- d. With your partner, write a formula that can be used to find the measure of an exterior angle of two secant lines.



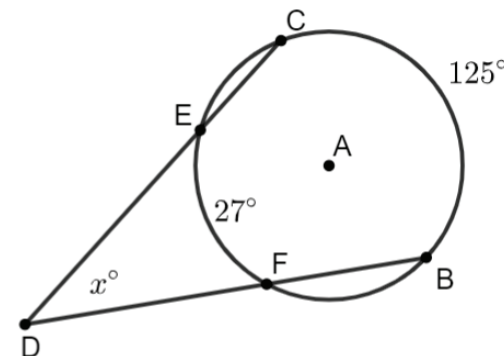
Exterior Secant Angle Theorem

The measure of an angle formed by two secants intersecting in the exterior of a circle is one-half the difference of the measures of the intercepted arcs.



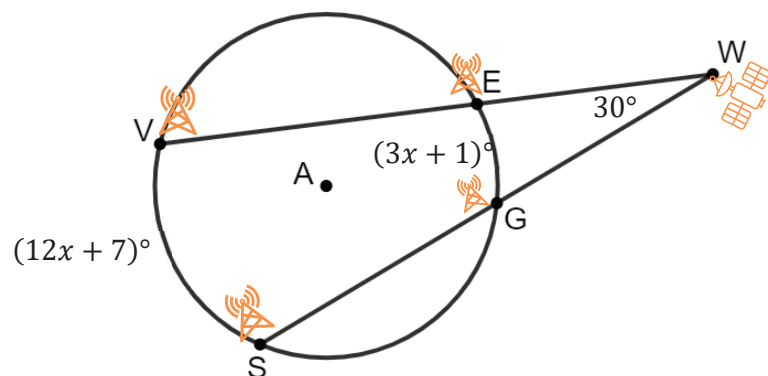
10.8.3 Guided Practice: Solving Problems Involving Angles Formed by Secants

1. Circle A has points C , E , F , and B on the circle. Secants $\overline{CD}$ and $\overline{BD}$ intersect at point D outside the circle. $m\widehat{CB} = 125^\circ$, $m\widehat{EF} = 27^\circ$, and $m\angle CDB = x^\circ$. What is the value of x ?



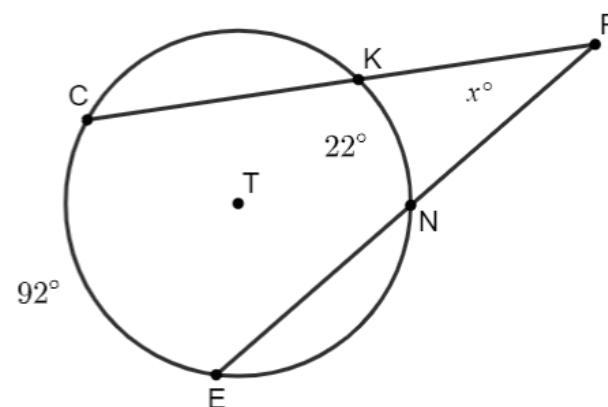
2. A satellite, point W , sends signals $\overline{VW}$ and $\overline{SW}$ to Earth. Signal $\overline{VW}$ hits tower E before reaching tower V . Similarly, signal $\overline{SW}$ hits tower G before reaching tower S . Scientists are interested in finding the measure of the intersecting arcs created by the two signals. They know $m\angle W = 30^\circ$, $m\widehat{VS} = (12x + 7)^\circ$, and $m\widehat{EG} = (3x + 1)^\circ$.

Find $m\widehat{VS}$ and $m\widehat{EG}$.



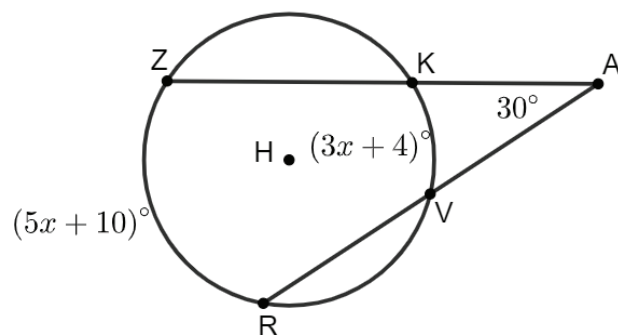
10.8.4 Your Turn!

1. Circle T has points C , K , N , and E on the circle. Secants $\overline{CR}$ and $\overline{ER}$ intersect at point R outside the circle. The $m\widehat{CE} = 92^\circ$, $m\widehat{KN} = 22^\circ$, and $m\angle CRE = x^\circ$. What is the value of x ?



2. Circle H has points Z , K , V , and R on the circle. Secants $\overline{ZA}$ and $\overline{RA}$ intersect at Point A outside the circle. The $m\widehat{ZR} = (5x + 10)^\circ$, $m\widehat{KV} = (3x + 4)^\circ$, and $m\angle ZAR = 30^\circ$.

Find the measure of $\widehat{ZR}$.



10.8.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class summarizing today's lesson.